

# CUIWEN ZHU

MC 5474 ♦ f3zhu@uwaterloo.ca

## EDUCATION

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**University of Waterloo** 2026 - present  
MMath - Combinatorics & Optimization *under co-supervision of Professor Chris Godsil and Professor Jason Bell*  
**University of Waterloo** 2022 - 2026  
Bachelor of Mathematics, Honours Pure Mathematics - Combinatorics & Optimization Minor  
(Dean's Honours)

## EXPERIENCE

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**Undergraduate Research Assistant**, University of Waterloo 2026/05 - 2026/08  
*MURA funded research with Professor Chris Godsil*  
Research on the continuous quantum walk, especially on circles with the uniform mixing property.  
**Part-time Undergraduate Research Assistant**, University of Waterloo 2025/10 - 2026/04  
*part-time research with Professor Chris Godsil*  
Research on the continuous quantum walk and algebraic graph theory.  
**Undergraduate Research Assistant**, University of Waterloo 2025/05 - 2025/08  
*MURA funded research with Professor Yu-Ru Liu*  
Research on number theory, especially on the Hardy–Ramanujan–Littlewood circle method, with application on Waring's Problem, Hasse Principle, and smooth numbers.  
**Part-time Algebra Tutoring**, Tutoring Centre, University of Waterloo 2024/05 - 2026/04  
Tutoring first and second year's algebra courses.  
**Undergraduate Research Assistant**, University of Waterloo 2024/01 - 2024/04  
*MURA funded research with Professor Elana Kalashnikov*  
Research on algebraic geometry, especially on the coordinate rings of Kronecker quiver moduli spaces.

## PROJECTS AND TALKS

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**“Bypassing the Exact: An Introduction to Quantum Walks and Their Applications”** 2026/06/25  
*Talk at Seminars in Canadian Undergraduate Mathematics Conference in the McMaster University*  
**“The Hasse Principle for Diagonal Forms via the Circle Method”** 2025/07/30  
*Talk at Student Number Theory Seminar in the University of Waterloo*  
**“Circle method on Waring Problem and Hasse Principle”** 2025/06/25  
*Talk at Seminars in Canadian Undergraduate Mathematics Conference in the University of Waterloo*  
**“Quiver Semi-invariants and SAGBI Bases”** 2024/07/09  
*Talk at Seminars in Canadian Undergraduate Mathematics Conference in the University of British Columbia*

## (CROSSLISTED-)GRADUATE COURSES TAKEN

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**CO 434:** Combinatorial Designs  
**CO 444:** Algebraic Graph Theory  
**PMATH 441:** Algebraic Number Theory  
**PMATH 445:** Representation of Finite Groups  
**PMATH 446:** Introduction to Commutative Algebra  
**PMATH 450:** Measure Theory

**PMATH 465:** Smooth Manifolds  
**PMATH 863:** Introduction to Lie Groups and Lie Algebras  
**PMATH 945:** Topics in Algebra; topic: Rings and their applications  
**PMATH 945:** Topics in Algebra; topic: Category Theory and Homological Algebra

**AWARDS**

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| President’s Research Award               | 2026 |
| Mathematics Undergraduate Research Award | 2026 |
| Mathematics Undergraduate Research Award | 2025 |
| President’s Research Award               | 2024 |
| Mathematics Undergraduate Research Award | 2024 |

**PUBLICATIONS AND PREPRINTS**

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| Uniform mixing and $\epsilon$ -uniform mixing on cycles | arXiv 2607.04207 |
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